

Enhancing Grammatical Error Correction: A Comparative Study of T5, BART, and System Combination Methods

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Abstract

Grammatical Error Correction (GEC) is a challenging sequence-to-sequence task focused on correcting writing errors, including grammar, punctuation and spelling errors. This paper investigates the effectiveness of pre-trained encoder-decoder models by fine-tuning and ensembling models. We fine-tune and compare four model variants—bart-base, bart-large, t5-base, and flan-t5-base—on benchmark datasets and evaluate them using standard shared task scorers. Overall, the T5 variants achieve stronger performance than the BART models, with flan-t5-base obtaining the highest $F_{0.5}$ score on CoNLL-2014. We further apply the Easy System Combination (ESC) method to merge edits from t5-base and bart-base. ESC achieves our best result on BEA-2019 with $F_{0.5}$ score of 73.61, although it does not outperform flan-t5-base on CoNLL-2014. These findings highlight that T5 architectures are highly effective for GEC and that system combination remains a powerful technique for performance gains.

1 Introduction

Grammatical Error Correction (GEC) is essential in writing assistance tools, yet producing accurate and fluent corrections remains challenging due to diverse learner errors and limited annotated data. Although Transformer-based models dominate modern GEC, the effects of architecture, model size, instruction tuning, and post-training system combination remain underexplored.

In this work, we compare four Transformer variants—t5-base, flan-t5-base, bart-base, and bart-large—using a consistent two-stage fine-tuning pipeline, and evaluate the Easy System Combination (ESC) framework for merging edits from multiple models without additional training. Our contributions include fine-tuning and evaluating these models on standard GEC corpora, applying

ESC as a post-training combination method, and providing empirical insights into how architectural choices, scale, instruction tuning, and system combination influence GEC performance.

2 Related Work

The CoNLL-2014 Shared Task (Ng et al., 2014) established one of the earliest standardized benchmarks for Grammatical Error Correction (GEC), where systems were dominated by rule-based and statistical machine translation (SMT) approaches. The task adopted the $F_{0.5}$ metric to emphasize precision.

The BEA-2019 Shared Task (Bryant et al., 2019) expanded the benchmark with the Write & Improve + LOCNESS corpus and introduced the ERRANT scorer (Bryant et al., 2017), marking the shift from SMT to Transformer-based models such as BERT, BART, and T5.

Building on this transition, we fine-tune T5 and BART models using a two-stage curriculum—broad adaptation on Lang-8 followed by specialization on high-quality corpora (NUCLE, FCE, W&I + LOCNESS). We also incorporate instruction-tuned Flan-T5 with task-specific prefixes and evaluate Easy System Combination (ESC) as a lightweight post-training method for combining model outputs.

3 Methodology

3.1 Core Model Architecture

For our Grammatical Error Correction (GEC) project, we are evaluating and comparing several state-of-the-art transformer architectures. Our investigation is centered on two main families of models: T5 (Raffel et al., 2020) and BART (Lewis et al., 2020)

3.1.1 T5 - Text-to-Text Transfer Transformer

The foundation of our first model is **T5 (Text-to-Text Transfer Transformer)**, developed by

Google. T5’s core innovation is its unified framework, which reframes all Natural Language Processing (NLP) tasks into a single text-to-text format. This makes it exceptionally well-suited for sequence-to-sequence tasks like GEC, where an input sequence (incorrect text) is directly mapped to an output sequence (corrected text). For this project, we are using the t5-base variant.

3.1.2 Flan-T5: The Instruction-Tuned Variant

Building on T5, we also implement Flan-T5, which differs not in architecture but in its training methodology (BytePlus, 2024). Flan-T5 is an instruction-tuned version of T5, meaning it underwent an additional fine-tuning phase on over 1,000 diverse NLP tasks formatted as natural language instructions.

We leverage this capability by prefixing every input sentence with the instruction: Correct the grammatical error in this sentence: . This prompts the model to perform a task it already conceptually understands, reframing GEC as a conditional generation problem that Flan-T5 is inherently designed to handle.

3.1.3 BART - Bidirectional and Auto Regressive Transformers

The foundation of our second model is **BART (Bidirectional and Auto Regressive Transformers)**, developed by Facebook AI. BART’s key innovation lies in its ability to combine bidirectional encoding, as seen in BERT, with autoregressive decoding, similar to GPT. This design allows the model to understand the full sentence context while generating fluent and grammatically corrected output.

For this project, we are using two variants of the model: **bart-base** and **bart-large**.

3.2 Experimental Setup

To ensure a fair comparison, all models were trained using a unified pipeline.

3.2.1 Data and Pre-processing

We use parallel source (.src) and target (.tgt) files. A common preprocessing pipeline tokenizes inputs and targets, which are truncated or padded to a 128-token sequence length. The tokenized target sequences are used as training labels.

The only model-specific difference was the use of a TASK_PREFIX ("Fix grammar: ") for the t5-base and TASK_PREFIX ("Correct the grammatical error in this sentence: ") for the flan-t5-base

models, which aligns with their instruction-tuned nature. This prefix was omitted for the bart-base and bart-large models.

3.2.2 Two-Stage Fine-Tuning Strategy

We employ a two-stage fine-tuning curriculum to maximize performance and model generalization.

Stage 1: General Domain Adaptation. Models are first fine-tuned on the large-scale Lang-8 corpus (lang8.train.auto.bea19). This stage adapts the models from their general-purpose pre-training to the broad domain of grammatical error correction (GEC), exposing them to a high volume and variety of learner errors.

Stage 2: Specialization. The checkpoints from Stage 1 are subsequently fine-tuned on a smaller, high-quality “gold-standard” dataset composed of the NUCLE, FCE, and WI+Locness corpora. This stage refines the model’s precision on expert-annotated, formal text.

3.2.3 Training Details and Hyperparameters

All models were trained using the Hugging Face Trainer with the AdamW optimizer. Models were evaluated every 500 steps, and the checkpoint with the lowest validation loss was saved.

A critical finding was the numerical instability of flan-t5-base when trained with 16-bit mixed precision (fp16). To ensure stable training, flan-t5-base was trained in full 32-bit precision (fp32).

Due to the size of bart-large, we used gradient accumulation to maintain an effective batch size of 16 while fitting the model in memory. All key hyperparameters are detailed in Table 1.

3.3 System Combination (ESC)

To leverage the diverse error correction capabilities of our models, we implemented a simple, post-processing system combination. Specifically, we applied the edits from our two best-performing base models (t5-base and bart-base) to the source text. To keep the approach manageable in complexity and straightforward to analyze, we limited the combination to these two models. This approach follows the ESC (Edit-based System Combination) method (Qorib et al., 2022), which formulates system combination for grammatical error correction as a binary classification task (Qorib et al., 2022). We trained the model on the base systems’ output from the BEA-2019 shared task development data,

Table 1: Hyperparameter settings for all models across both fine-tuning stages.

Hyperparameter	t5-base	flan-t5-base	bart-base	bart-large
<i>Stage 1 (General Adaptation)</i>				
Learning Rate	5e-5	5e-5	5e-5	5e-5
Warmup Steps	10,000	10,000	10,000	10,000
Epochs	2	2	2	2
Batch Size (per device)	16	16	16	8
Grad. Accum. Steps	1	1	1	2
Precision (fp16)	True	False	True	True
<i>Stage 2 (Specialization)</i>				
Learning Rate	1e-5	1e-5	1e-5	1e-5
Warmup Steps	100	500	500	500
Epochs	10	5	5	5
Batch Size (per device)	16	16	16	8
Grad. Accum. Steps	1	1	1	2
Precision (fp16)	True	False	True	True

successfully leveraging model diversity to produce improved corrections.

4 Results and Analysis

We evaluate all models on the CoNLL-2014 test set (M2 Scorer) and the BEA-2019 test set (ERRANT Scorer).

4.1 Main Results

Table 2 presents the primary results (precision, recall and $F_{0.5}$) of our experiments. We find that the t5-base model, trained with our two-stage strategy, achieves the highest $F_{0.5}$ score on both test sets.

4.2 Analysis of Model Performance

Our results in Table 2 lead to several key insights:

Flan-T5 vs. t5-base: The instruction-tuned flan-t5-base achieves the highest $F_{0.5}$ score on CoNLL-2014 (67.45) by sacrificing recall for a significant precision boost. However, t5-base is the stronger all-rounder, achieving a better $F_{0.5}$ score on the BEA-2019 benchmark (68.97). We hypothesize this is due to flan-t5-base’s training instability (fp32) and its "generalist" nature from being pre-trained on 1,000+ tasks, which made it less suited to our aggressive GEC-only specialization.

T5-family vs. BART-family: The T5 models consistently outperform the BART models in $F_{0.5}$ score. We attribute this to T5’s "span-infilling" pre-training objective, which is a more direct fit for the GEC task of replacing incorrect text spans. On BEA-2019, t5-base (68.97) beats even the much larger bart-large (67.20), supporting our hypothesis that the larger model overfit the small Stage

2 dataset. A key strategic difference is apparent: BART models are recall-oriented (with bart-large achieving the highest recall of 63.08%), while T5 models are precision-focused (with flan-t5-base reaching 78.90% precision).

System Combination (ESC): Our ESC implementation yielded a significant and split result. On the BEA-2019 test set, it was the clear winner, achieving an $F_{0.5}$ of **73.61**—a 4.64-point gain over the best single model. It successfully combined the complementary strengths of the precision-focused t5-base and the recall-oriented bart-base, achieving the highest precision of all (79.93%). Conversely, on CoNLL-2014, the ESC implementation failed, scoring 65.12, which is lower than both T5 models. This suggests that the two test sets have different error distributions. On CoNLL-14, the models’ edits likely had more negative interactions (error amplification), which our ESC model was not trained to resolve.

Our hypothesis on CoNLL-14 test set failure: Combining only a small number of base models can limit the performance gains achievable by system combination methods like ESC. As demonstrated in the paper (Qorib et al., 2022), adding more diverse base systems consistently improves the $F_{0.5}$ score of the combined model even when individual systems are weaker. Thus, the modest result on CoNLL-2014 may in part reflect insufficient model diversity in our ensemble.

4.3 Ablation: Impact of Two-Stage Training

To justify our complex two-stage methodology, we performed an ablation study. We hypothesized that Stage 2 specialization in gold-standard data would be critical for performance. To challenge this, we

Table 2: Main GEC performance on the CoNLL-2014 and BEA-2019 test sets. All models are from the final Stage 2 specialization. The best $F_{0.5}$ score is in **bold**.

Model	CoNLL-2014 (M2 Scorer)			BEA-2019 (ERRANT Scorer)		
	P	R	$F_{0.5}$	P	R	$F_{0.5}$
t5-base	74.40	45.65	66.08	70.72	62.74	68.97
flan-t5-base	78.90	42.03	67.45	70.21	54.72	66.75
bart-base	65.43	50.14	61.67	64.19	61.08	63.54
bart-large	71.02	51.15	65.90	68.34	63.08	67.20
ESC(t5-base+bart-base)	77.62	39.62	65.12	79.93	55.93	73.61

compared our full pipeline against a simpler single-stage training on our bart-base and bart-large models. This single-stage approach combined all our datasets on hand (FCE, NUCLE, W&I + Locness, and the Lang-8 Corpus) into one training run. The rationale was to test if simply allowing exposure to all diverse datasets at once would allow the model to generalize effectively, potentially matching or exceeding our more complex two-stage curriculum.

The results for bart-base and bart-large, are shown in Table 3.

Table 3: Ablation study on the BEA-2019 test set, comparing single-stage vs. two-stage training for bart-base.

Training Strategy	P	R	$F_{0.5}$
bart-base (Stage 1 Only)	68.71	48.89	63.56
bart-base (Stage 1 + 2)	64.19	61.08	63.54
bart-large (Stage 1 Only)	66.27	43.65	60.04
bart-large (Stage 1 + 2)	68.34	63.08	67.20

4.3.1 Analysis of Training Strategy

Our ablation study reveals that the impact of the two-stage curriculum is highly dependent on model scale.

As shown in Table 3, for bart-base, the addition of Stage 2 specialization provided no performance gain. The $F_{0.5}$ scores were nearly identical (63.56 vs 63.54).

In contrast, for bart-large, the two-stage strategy was critically important, producing a massive **+7.16** $F_{0.5}$ gain (60.04 $\rightarrow$ 67.20). The subsequent fine-tuning on the small, high-quality Stage 2 data was essential for the model to converge and learn the GEC task effectively.

This result suggests that our two-stage curriculum is a highly effective strategy for training larger-capacity models, while smaller models may be better suited to a single-stage training on large, diverse data.

5 Limitations and Future Work

Despite the strengths of our approach, several limitations remain. First, our experiments focus mainly on base-sized models, with only bart-large providing a comparison point. Hardware constraints prevented us from evaluating larger variants, such as T5-large or Flan-T5-XL, which may yield further improvements.

Second, while our two-stage fine-tuning strategy was applied to all models, our ablation study (Section 4.3) was limited to the BART architecture. We did not compare against a single-stage T5 model; therefore, while our two-stage T5 models performed well, we cannot definitively conclude that this curriculum is the optimal strategy for T5, and this remains an area for future work.

Third, our successful ESC implementation on BEA-2019 failed on CoNLL-2014. This highlights its sensitivity to the test data distribution and our simple combination rule. Future work should explore more sophisticated, learned combination methods to predict edit quality, as well as per-error-type analysis to further probe the complementary nature of T5 and BART.

6 Conclusion

In this paper, we presented a systematic comparison of T5 and BART models for GEC.

Our results show that T5-family models, particularly t5-base, are the strongest single-model performers. Most significantly, we show that a simple Easy System Combination (ESC) of the precision-focused t5-base and the recall-oriented bart-base achieves a state-of-the-art $F_{0.5}$ score of **73.61** on the BEA-2019 test set. This confirms that combining models with complementary strengths remains a highly effective strategy for advancing GEC performance.

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